# Linear Regression - Complete Notes

## 1. Introduction

Linear Regression is one of the most fundamental supervised machine learning algorithms. It is used to model the relationship between dependent and independent variables by fitting a linear equation to observed data.

## 2. Types of Linear Regression

**Simple Linear Regression:** Uses one independent variable.
**Multiple Linear Regression:** Uses multiple independent variables.

## 3. Equation of Linear Regression

Simple Linear Regression equation:

**$y = mx + c$**

Where:
$y$ = dependent variable
$x$ = independent variable
$m$ = slope
$c$ = intercept

## 4. Objective

The main objective is to find the best-fit line that minimizes prediction error.

## 5. Cost Function

The commonly used cost function is Mean Squared Error (MSE):

$MSE = (1/n) * \Sigma(y_{actual} - y_{predicted})^2$

## 6. Gradient Descent

Gradient Descent is an optimization algorithm used to minimize the cost function by updating parameters iteratively.

## 7. Assumptions of Linear Regression

• Linear relationship
• Independence of observations
• Homoscedasticity
• Normal distribution of residuals
• No multicollinearity

## 8. Evaluation Metrics

• Mean Absolute Error (MAE)
• Mean Squared Error (MSE)
• Root Mean Squared Error (RMSE)
• $R^2$ Score

## 9. Advantages

• Easy to implement
• Interpretable results
• Fast training

## 10. Disadvantages

• Sensitive to outliers
• Assumes linearity
• Poor performance on complex datasets

## 11. Applications

• House price prediction
• Stock market prediction
• Sales forecasting
• Risk assessment

## 12. Python Example

```
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(X_train, y_train)
pred = model.predict(X_test)
```

## Evaluation Metrics Table

| **Metric** | **Meaning** |
| --- | --- |
| MAE | Average absolute error |
| MSE | Average squared error |
| RMSE | Square root of MSE |
| $R^2$ Score | Goodness of fit |